RAG CHATBOT TECHNICAL REPORT

AMLGO LABS JUNIOR AI ENGINEER ASSIGNMENT

Aarav | April 2026

Document Structure and Chunking logic

The source document is the eBay User Agreement — a dense legal document spanning 20 pages and 68,253 characters. The document was loaded using the pypdf library which extracted clean, readable text requiring no additional cleaning or HTML removal.

The text was split into 89 overlapping chunks using a custom sentence-aware chunking function. Each chunk targets approximately 1,000 characters (roughly 150-200 words), which falls within the assignment's 100-300 word range and was chosen specifically for legal text density — shorter chunks risked cutting through multi-clause sentences.

Because legal documents tend to have longer sentences and words so a chunk overlap of 90 characters was used to ensure sentences spanning chunk boundaries were not lost, which is especially important in legal documents where a single clause can span multiple lines.

Embedding Model and Vector Database

Embedding Model: all-MiniLM-L6-v2 from the sentence-transformers library was selected for its balance of speed and semantic quality. It produces 384-dimensional dense vector embeddings and runs efficiently on CPU with no GPU required. The model is only 90MB making it practical for local deployment.

Vector Database: FAISS (Facebook AI Similarity Search) was chosen for its simplicity and efficiency. It requires no server setup, stores the entire index in a single file (faiss_index.index), and performs fast L2 nearest-neighbor search across all 89 chunk vectors. The index is loaded once at app startup and cached for the session.

Prompt Format and Generation Logic

The following is the exact prompt template implemented in src/generator.py:

```
prompt = f"""You are a friendly and helpful assistant specializing in eBay's User Agreement.


You have two modes:

1. If the user is making small talk or greeting you, respond in a warm, friendly and conversational way. Keep it brief and invite them to ask about the eBay User Agreement.

2. If the user is asking a question related to eBay, answer based ONLY on the provided context below. If the answer is not in the context, say "I couldn't find specific information about that in the eBay User Agreement."


Always be warm, clear and concise. When the context contains the answer, be direct and specific. Do not hedge or redirect unnecessarily.


CONTEXT:

{context}


USER QUESTION:

{query}


ANSWER:"""
```

The {context} variable contains the top 5 retrieved chunks injected as numbered source passages, and {query} contains the user's original question.

The LLM used is LLaMA 3.3 70B served via the Groq API, which provides extremely fast inference at no cost. At query time, the top 5 most semantically similar chunks are retrieved from FAISS and injected into a structured prompt template alongside the user query.

The prompt is composed of four distinct components:

- *System Persona:* The model is instructed to act as a friendly and helpful assistant specializing in the eBay User Agreement. This sets the tone and scope of the model's behavior, preventing it from answering questions outside the domain of the document.
- *Behavioral Modes:* ***The prompt defines two explicit modes of operation.*** In small talk mode, the model responds warmly and conversationally when the user is greeting or chatting, inviting them to ask a document-related question. In Q&A mode, the model answers strictly from the retrieved context. ***This dual-mode design improves user experience without sacrificing grounding.***
- *Context Injection:* The top 5 retrieved chunks are injected into the prompt under a clearly labeled CONTEXT section. This structured format helps the model attribute its answer to specific sections of the document rather than blending them arbitrarily.
- *Grounding Instruction:* The model is explicitly told to answer only from the provided context and to acknowledge when information is not available rather than hallucinate. This is the most critical component for factual accuracy in a RAG system. Additionally, the model is instructed to be direct and specific when the answer is present, and to avoid unnecessary hedging.
- *Generation Parameters:* Temperature is set to 0.1 to minimize creative deviation and keep responses factual. Max tokens is capped at 1024 to prevent unnecessarily long responses. Responses are streamed token by token back to the Streamlit interface using Groq's streaming API, creating a real-time typing effect in the chat UI.

Example Queries and Responses

Q1 What happens if eBay suspends my account? — Success

If eBay suspends your account, they may limit, restrict, or downgrade your seller account, delay or remove hosted content, remove any special status associated with your account, remove or demote listings, reduce or eliminate discounts, and take technical and/or legal steps to prevent you from using their Services. Additionally, you may be subject to fees and recovery of expenses for policy monitoring and enforcement.

Analysis: Strong success case. The retriever correctly identified the policy enforcement section and the model produced a detailed, accurate, multi-point answer.

Q2 How does eBay handle disputes between buyers and sellers? — Failure

eBay facilitates the resolution of disputes between buyers and sellers through various programs. However, eBay has no control over and does not guarantee the existence, quality, safety, or legality of items advertised, the truth or accuracy of users' content or listings, or that a buyer or seller will actually complete a transaction or return an item.

Analysis: Failure case. The query term "disputes" semantically matched eBay's general disclaimer section instead of the Legal Disputes section which covers arbitration and NAM procedures. This highlights a known RAG limitation — semantically similar vocabulary does not always map to the most relevant document section. A hybrid keyword + semantic search approach could improve this.

Note: ***When Framed differently, the chatbot seems to understand the question better, as displayed in the screenshot at the end of the document.***

Q3 Can eBay change its policies without notice? — Success

According to the eBay User Agreement, eBay can change its selling fees without advance notice for temporary promotions or changes that result in the reduction of fees. However, for other changes, they will provide 30 days' notice by posting the amended terms on www.eBay.com, and also notify users through the eBay Message Center and/or by email.

Analysis: Strong success case. The model correctly distinguished between two different notice periods in the document — 14 days for fee changes and 30 days for policy amendments — demonstrating nuanced retrieval and accurate synthesis.

Q4 What rights do I give up when I agree to eBay's User Agreement? —Partial Success

According to the User Agreement, when you agree to it, you expressly waive any protections that would otherwise limit the coverage of the release to include only those claims which you may know or suspect to exist in your favor at the time of agreeing to this release. This means you give up certain rights and protections, including the right to bring certain claims against eBay.

Analysis: Partial success. The answer is technically correct but incomplete — the document also covers waiver of jury trial rights, class action rights, and individual court claims in the arbitration section. The retriever found the release clause but missed the arbitration waiver section, showing that broad questions spanning multiple document sections can result in incomplete answers.

Q5 What is the process to opt out of arbitration? — Success

To opt out of arbitration, you need to complete, sign, and mail the Opt-Out Notice form to eBay at the address provided: 583 West eBay Way, Draper, UT 84020, ATTN: LITIGATION DEPARTMENT, RE: OPT-OUT NOTICE. The Opt-Out Notice must be sent by you individually, and any opt-out notices from third parties will have no effect.

Analysis: Strong success case. The model retrieved the correct procedural section and provided specific, accurate details including the exact mailing address and department name.

Notes on Hallucination, Limitations, and Slow Responses

- ***Hallucination Control:*** Hallucination risk was minimized through two mechanisms — grounding responses strictly in retrieved chunks via the prompt template, and setting temperature to 0.1. In testing, the model consistently acknowledged when information was not available rather than fabricating answers.

- ***Retrieval Failures:*** As demonstrated in Q2 and Q4, semantic search occasionally retrieves the wrong document sections when query vocabulary is ambiguous or when the answer spans multiple sections. A hybrid retrieval approach combining semantic search with keyword matching (BM25) could improve accuracy for these cases.

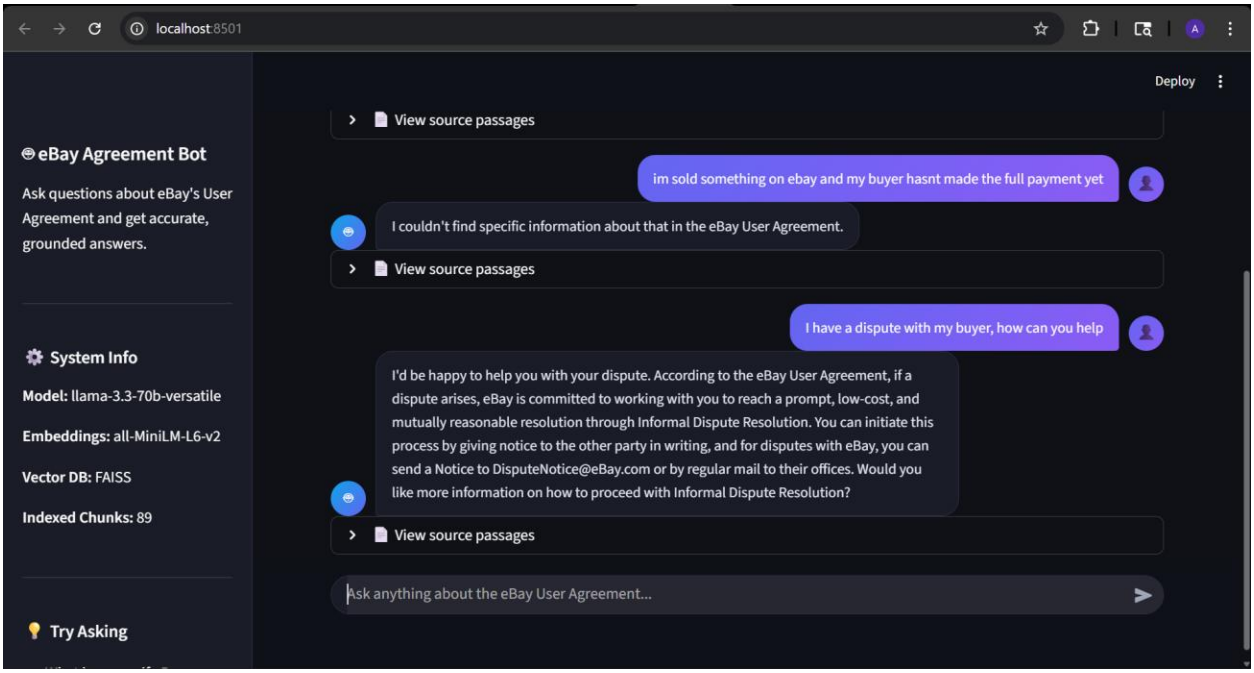
- ***Chunk Boundary Artifacts:*** The current chunking implementation splits text based on character count, which can result in incomplete sentences at chunk boundaries even with 90 character overlap. A more robust approach would be to use a sentence-aware splitter such as LangChain's RecursiveCharacterTextSplitter or spaCy's sentence segmentation, which splits on sentence boundaries first and only falls back to character limits when necessary. This would produce cleaner, more semantically complete chunks and improve retrieval quality. Additionally, producing a larger number of smaller, more granular chunks would increase the vector index size and allow FAISS to retrieve more targeted passages, leading to more precise and complete answers.

- ***External References:*** The eBay User Agreement frequently references external pages for specific details such as exact fee percentages and selling fee schedules. Since these external pages are not part of the indexed document, questions requiring that information cannot be fully answered.

- ***Response Speed:*** The first query has a slight delay due to the MiniLM embedding model loading into memory on CPU. Subsequent queries are fast since the model and FAISS index

are cached in Streamlit's session state. Groq's API inference is extremely fast even for the 70B model.

- ***Frontend Limitations:*** The chatbot interface was built using Streamlit, which is well suited for rapid AI application prototyping but has inherent limitations as a frontend framework. Streamlit re-renders the entire page on every interaction, making smooth animations and dynamic UI transitions difficult to implement. A production-grade interface built with React and a proper CSS framework would allow for a significantly more intuitive and polished user experience, including smooth message animations, better mobile responsiveness, and more flexible layout control.

- ***Model Limitations:*** LLaMA 3.3 70B via Groq occasionally adds conversational filler when the retrieved context is sparse. This was partially mitigated through prompt engineering but still remains a minor issue for ambiguous queries.

Chatbot Screenshot